



*Barium*  
*Calcium*  
*Lithium*  
*Potassium*  
*Sodium*

## Precise determination of potassium and sodium via BWB Flame Photometer

A reliable technique for quality control

**F**lame photometry is a technique for measuring the concentration of certain elements based on the intensity of light they emit when excited in a flame. Elements suitable for this analysis must ionize readily in aqueous solutions and have low excitation energies. This makes Group 1 (alkali metals) and Group 2 (alkaline earth metals) ideal candidates. Among these, potassium (K) and sodium (Na) are particularly important for various industries such as pharmaceuticals, food and beverage production, and environmental monitoring. The BWB Technologies Flame Photometer is a state-of-the-art instrument designed for precise and consistent measurement of these elements.

### BWB Flame Photometer Features

The BWB Flame Photometer offers several key features for efficient and reliable analysis:

- **Multi-channel detection:** Enables simultaneous measurement of multiple elements, improving analysis speed.
- **User-friendly interface:** Simplifies operation and data management on the BWB Flame Photometer.

- **Advanced calibration:** Supports both single-point and multi-point/multi-ion calibration for increased accuracy on the BWB instrument.
- **High sensitivity:** Detects low concentrations of elements, ensuring wider applicability for the BWB Flame Photometer.
- **Built-in air compressor:** Reduces noise and minimizes lab space requirements for the BWB instrument.
- **Optional automatic fluid handling system:** Automates analysis and increases lab productivity with the BWB Flame Photometer.
- **Access control technology:** Ensures compliance with Good Laboratory Practices (GLP), Laboratory Information Management Systems (LIMS), and multi-user access features for the BWB Flame Photometer.

### Methodology

**Instrument startup and stabilization:** The BWB Flame Photometer is powered on, and the flame ignited. It is allowed to stabilize for 45 minutes to ensure optimal performance.

**Standard preparation:** A series of standards is prepared from a 10ppm stock solution of sodium and

# Application showcase

potassium. Concentrations ranging from a blank (0 ppm) to 0.10 ppm are created in increments of 0.02 ppm using a dilution process. Deionized water is used for dilution.

**Calibration:** The standards are aspirated into the BWB Flame Photometer in ascending concentration order for multi-point/multi-ion calibration.

**Sample analysis:** The instrument is switched to read mode, and the sample solution is aspirated for measurement using the BWB Flame Photometer.

## Considerations for low concentration analysis

For samples with low sodium and potassium concentrations, diluting both the sample and standards with up to 20% xylene can theoretically improve signal strength by raising the flame temperature in the BWB Flame Photometer. However, this dilution needs to be factored into the results as it reduces the water content and may push the sample below the instrument's detection limit. Further experimentation is recommended to determine the optimal water-to-xylene ratio for optimal signal enhancement without compromising accuracy when using the BWB Flame Photometer.

## Benefits of BWB Technologies Flame Photometer

**High precision and accuracy:** The BWB Flame Photometer delivers reliable measurements for quality control and regulatory compliance.

**Efficiency:** Automated features and multi-channel detection on the BWB Flame Photometer increase sample throughput and reduce analysis time.

**User-friendliness:** The BWB Flame Photometer boasts an intuitive interface and straightforward procedures, simplifying training and operation.

**Versatility:** The BWB Flame Photometer is suitable for a wide range of applications beyond potassium and sodium analysis.

## Conclusion

The BWB Technologies Flame Photometer is a valuable tool for laboratories requiring precise and efficient analysis of potassium and sodium. Its advanced features and reliable performance make it ideal for ensuring product quality and compliance across various industries. By following the outlined procedures, users can achieve accurate and reproducible results, enhancing their analytical capabilities with the BWB Flame Photometer.



The BWB XP Flame Photometer

## Procedure

